Fiscal Year:	FY 2011	**Task Last Updated:**	FY 12/02/2010
PI Name:	Burma, Sandeep Ph.D.		
Project Title:	Radiation and Gliomagenesis: A Sensitive Model System to Evaluate the Tumorigenic Potential of HZE Particles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	78229	**Congressional District:**	21
Comments:	NOTE: Formerly at University of Texas Southwestern Medical Center at Dallas until fall 2019.		
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Radiobiology NNJ09ZSA001N
Start Date:	02/01/2010	**End Date:**	01/31/2013
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bachoo, Robert (University of Texas Southwestern Medical Center) Boothman, David (University of Texas Southwestern Medical Center)		
Grant/Contract No.:	NNX10AE08G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	There is a paucity of systematic studies carried out with well-defined model systems to evaluate the tumorigenic potential of HZE particles that are presumed to pose a significant cancer risk to astronauts. We have shown that these particles induce DNA double-strand breaks (DSBs) that are repaired slowly and incompletely and trigger persistent DNA damage signaling events. In order to evaluate the carcinogenic consequences of such unrepaired DNA lesions, we propose to utilize genetically engineered astrocytes/neural stem cells (NSCs) as well as transgenic mouse models representing progressive steps in the development of Glioblastoma Multiforme (GBM). These lethal brain tumors are the third leading cause of cancer-related death among adults aged 30 to 50 years that are known to be induced by ionizing radiation. The past year has seen unprecedented advances in the genomic analyses of adult GBM tumors by the Cancer Genome Atlas Network, which reveal that these tumors have radically altered genomes with 5 key genetic changes dominating: loss of Ink4a, Arf, p53, or PTEN and amplification of EGFR (especially the constitutively active EGFRvIII). We have developed astrocyte and NSC lines as well as mouse models with targeted deletions of these genes in logical combinations representing the progression of primary or secondary GBMs. This genetic progression model, though reductionist in approach, provides a realistic framework for evaluating and understanding how known genetic mutations may cooperate with charged particle radiation and contribute to the initiation and progression of high-grade gliomas. We hypothesize that poorly repaired DNA damage produced by charged particles will result in malignant progression towards brain cancer, and that this progression can be accelerated by predisposing oncogenic activations/tumor suppressor losses. More importantly, these "pre-initiated" murine astrocytes/NSCs will allow us to rapidly quantify the transforming potential of charged particles relative to gamma rays both in vitro and in vivo.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	This grant seeks to understand cancer risks (especially the risk of brain cancers) from particle radiation in outer space. However, charged particles are also being used in the clinic for targeted tumor therapy. In addition to evaluating cancer risks to astronauts, our project will also contribute significantly to our understanding of cancer cell killing by charged particles in the clinic and can shed light on the risk of secondary cancers from particle therapy.		
Task Progress:	Significance of funded research: Ions of high atomic number and energy (HZE particles) pose a significant cancer risk to astronauts on prolonged space missions. The decision to launch future long-duration space missions hinges, in part, upon the rapid estimation of cancer risks from these particles. Yet, there is a paucity of systematic studies carried out with well-defined model systems to evaluate the tumorigenic potential of HZE particles. In order to understand the carcinogenic consequences of heavy particle irradiation, we propose to utilize genetically engineered astrocytes/neural stem cells (NSCs) as well as transgenic mouse models representing progressive steps in the development of lethal brain tumors, Glioblastoma Multiforme (GBM). Our model systems provide a realistic framework for evaluating and understanding how known genetic mutations may cooperate with charged particle radiation and contribute to the initiation and progression of high-grade gliomas. With this backdrop, the specific aims of this project are: Aim 1: Evaluation of DNA damage responses to charged particles in neural stem cells (NSCs) and astrocytes in vitro and in vivo. Aim 2: Evaluation of the tumorigenic potential of charged particles relative to gamma rays in vitro and in vivo. Aim 3: Identification of the genetic and gene expression changes underlying particle-induced tumorigenesis. Brief summary of progress: In experiments carried out during the first year of the project, we have been able to evaluate the relative tumorigenic potential of heavy particle irradiation compared to gamma-rays using in vitro model systems. Our in vitro studies resulted in a manuscript that was accepted for publication in Carcinogenesis [see below]. As this is the first year of funding for this project, we anticipate completing the remaining proposed experiments in the coming years. Detailed summary of progress:		

	To determine whether Fe-induced DSBs are more potently tumorigenic than gamma ray-induced breaks, we irradiated “sensitized” murine astrocytes that were deficient in Ink4a and Arf tumor suppressors and injected the surviving cells sub-cutaneously into nude mice. Using this model system, we find that Fe ions are potently tumorigenic, generating tumors with significantly higher frequency and shorter latency compared to tumors generated by gamma rays. Tumor formation by Fe-irradiated cells is accompanied by rampant genomic instability and multiple genomic changes, the most significant of which is loss of the p15/Ink4b tumor suppressor due to deletion of a chromosomal region harboring the CDKN2A and CDKN2B loci. The additional loss of p15/Ink4b in tumors derived from cells that are already deficient in p16/Ink4a bolsters the hypothesis that p15 plays an important role in tumor suppression, especially in the absence of p16. Indeed, we find that re-expression of p15 in tumor-derived cells significantly attenuates the tumorigenic potential of these cells, indicating that p15 loss may be a critical event in tumorigenesis triggered by complex DSBs. Reference: Loss of p15/Ink4b accompanies tumorigenesis triggered by complex DNA double-strand breaks. C. V. Camacho, B. Mukherjee, B. McEllin, L-H. Ding, B. Hu, A. Habib, X-J. Xie, C. Nirodi, D. Saha, M. Story, A. Balajee, R. M. Bachoo, D. A. Boothman, and S. Burma. Carcinogenesis 31:1889-1896 (2010)
Bibliography Type:	Description: (Last Updated: 12/09/2025)
Abstracts for Journals and Proceedings	Camacho CV, Mukherjee B, McEllin B, Ding L-H, Hu B, Habib AA, Xie, X-J, Nirodi CS, Saha D, Story MD, Balajee AS, Bachoo RM, Boothman DA, Burma S. "Loss of p15/Ink4b accompanies tumorigenesis triggered by complex DNA double-strand breaks." Presented at the 56th Annual Meeting of the Radiation Research Society, Maui, Hawaii, September 25-29, 2010. Radiation Research Abstract Book, 56th Annual Meeting of the Radiation Research Society, 2010. http://www.abstractsonline.com/Plan/ViewAbstract.aspx?mID=2569&sKey=359561f3-493c-4d5c-b014-47e677719c70&cKey=f176ed84-f975-4f33-8030-365a07676c9c&mKey=5fb93787-2553-4b3b-a915-8c1ae71f127a , Sep-2010
Articles in Peer-reviewed Journals	Hatanpaa KJ, Burma S, Zhao D, Habib AA. "Epidermal growth factor receptor in glioma: signal transduction, neuropathology, imaging, and radioresistance." Neoplasia. 2010 Sep;12(9):675-84. PMID: 20824044 , Sep-2010
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Awards	Camacho CV, Mukherjee B, McEllin B, Ding L-H, Hu B, Habib AA, Xie, X-J, Nirodi CS, Saha D, Story MD, Balajee AS, Bachoo RM, Boothman DA, and Burma S, "2010 Best Student Poster Award for Camacho CV, Mukherjee B, McEllin B, Ding L-H, Hu B, Habib AA, Xie, X-J, Nirodi CS, Saha D, Story MD, Balajee AS, Bachoo RM, Boothman DA, and Burma S, 'Loss of p15/Ink4b accompanies tumorigenesis triggered by complex DNA double-strand breaks,' Enviromental Mutagen Society Annual Meeting, Ft Worth, TX, October 2010." Oct-2010
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Awards	McEllin B, Boies L, Camacho CV, Nannepaga S, Mashimo T, Vemireddy V, Sirasanagandla S, Bachoo RM, Burma S. "2nd Place Student Poster Presenation for 'Generation of a novel mouse model system to study HZE particle-induced gliomagenesis,' 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, NY, May 2010." May-2010